Research Statement

My research studies how to distinguish machine-generated text from human writing, and what that distinction reveals about both the models that produce the text and the domains it comes from. In my Master's thesis at the Institute for Natural Language Processing (IMS), University of Stuttgart, I take this question further, analysing the differences between AI- and human-produced paraphrases: which linguistic features act as reliable fingerprints of the generating source, and how those fingerprints shift under different prompting styles. These questions are what draw me to COMPASS, since they align closely with your group's concern with the misuse of generative models. They also connect directly to your work on tracing text provenance: where the Adversarial Watermarking Transformer (AWT) injects a traceable signal into generated text, I study the intrinsic signals that models already leave behind. Different as these two approaches are, both treat provenance and attribution as a defence against misuse.

My background combines linguistics and computer science, which is what allows me to work at this level of detail. I hold a B.A. in Linguistics and a Diploma in Software Development, and I am currently completing an M.Sc. in Computational Linguistics, with coursework in Machine Learning, Deep Learning, and Foundation Models. The combination of these disciplines shapes how I approach machine-generated text detection: I am interested not only in whether a classifier can separate AI-generated from human-written text, but also in which features it relies on, and in whether those features remain stable under paraphrasing and adversarial prompting.

My recent projects extend this focus on understanding and addressing the risks that generative models pose. In my ongoing work with the IRIS group (DANIS project), I study machine-generated text detection using informative, interpretable linguistic features rather than opaque classifiers, and I examine how reliably these features hold up across different domains and models. This work, recently submitted to ARR, is a step toward detectors whose decisions can be examined rather than trusted blindly. Earlier projects broadened this concern with harmful content: I worked on multilingual text detoxification (published at the BUCC workshop, LREC 2026) and on multimodal Arabic hate-meme detection (Mahed2025, ArabicNLP), both aimed at identifying and reducing harmful generated content. Different as these projects were in their specific tasks, together they taught me to treat safety as a measurable property of a system rather than as an afterthought.

Within your group, I would like to take this agenda in two directions. The first is robust provenance: understanding why detection signals are fragile under prompt variation, and whether interpretable, white-box features can be made robust to adversarial paraphrasing. The second direction is where your recent work on test awareness and safety alignment especially interests me, namely the finding that a model's behaviour shifts when it becomes aware of its context. My own research points in the same direction: prompt style appears to alter the linguistic fingerprint of generated text, which I see as one facet of this broader context-dependence. I am also eager to learn more about the agentic side of your research, which is newer to me, but where I expect provenance and interpretability to become increasingly important. As the quality of text generation improves, the line between human and machine text blurs and the intrinsic signals separating them grow fainter, which makes provenance and detection at once more difficult and more consequential.

I am applying for a PhD position because I want to do foundational, careful work on these questions, and I believe COMPASS and the wider Tübingen research environment are where I can pursue it most effectively. I would be honoured to contribute to your group's work on the safety and security of generative models. Thank you very much for considering my application.

Yassir El Attar